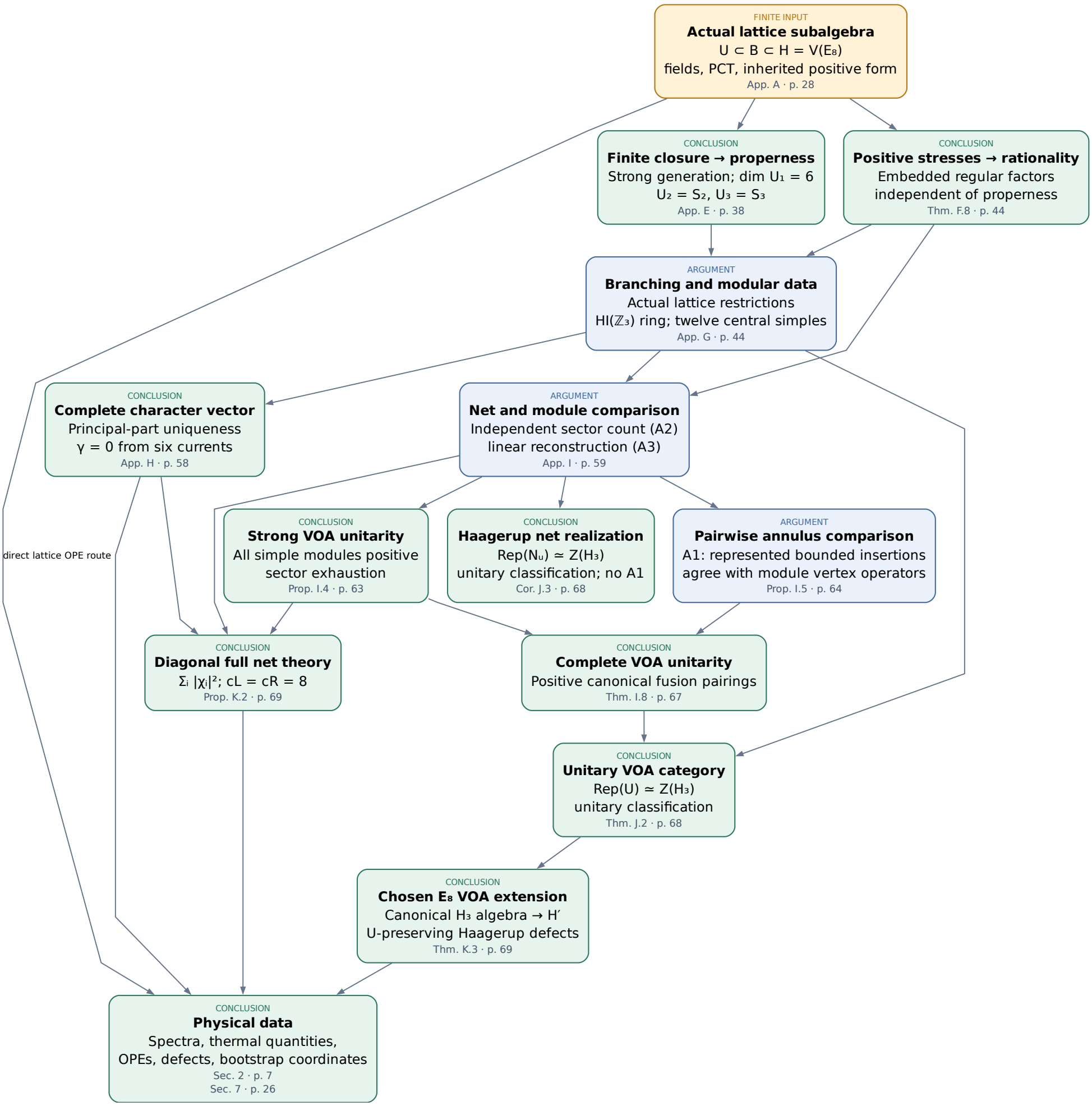
Argument overview

The construction, net realization, VOA realization, and observables have different dependencies.

Finite input | Argument in draft | Conclusion | External input | Optional route | Open boundary



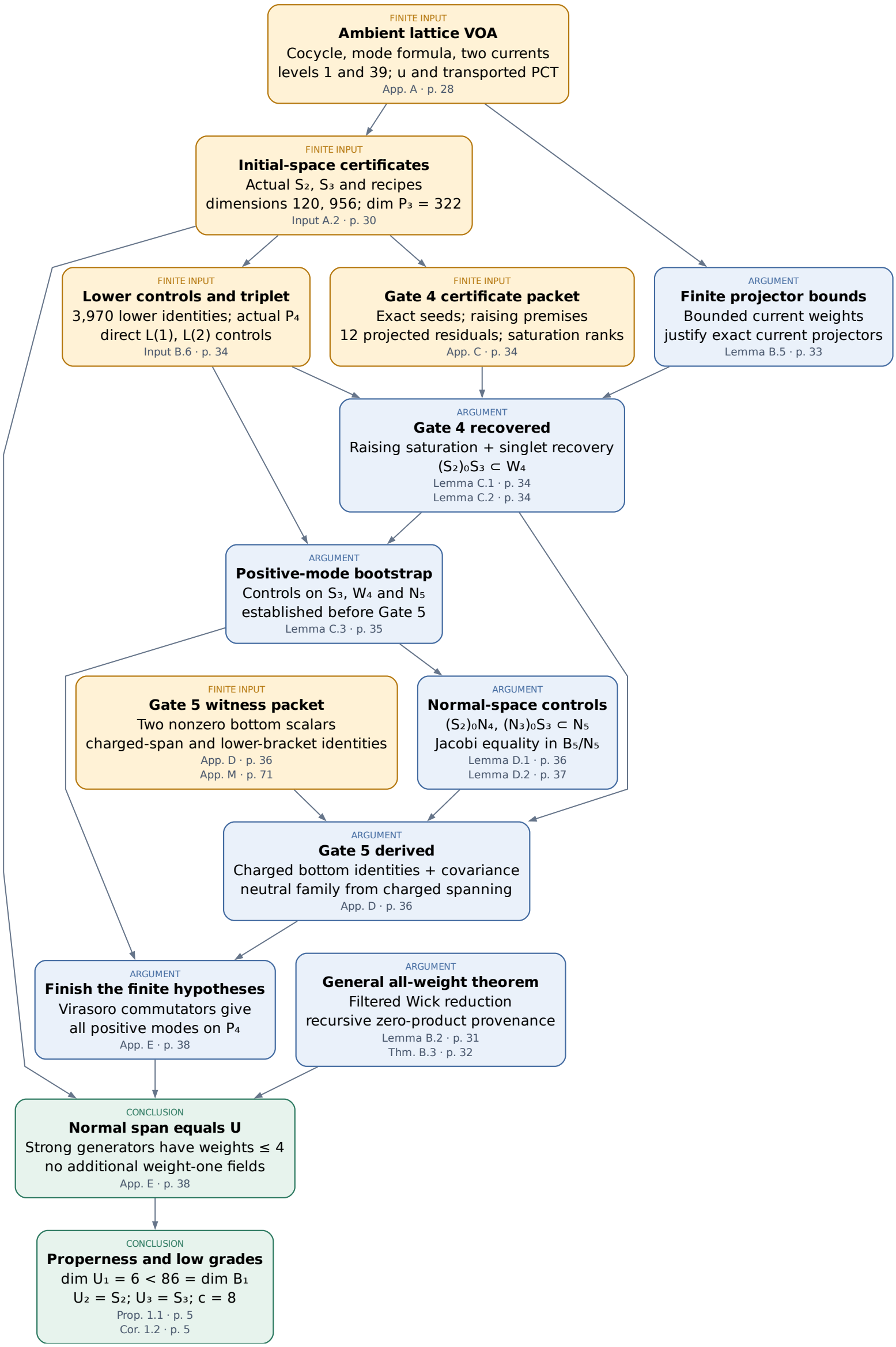
Arrows show selected logical dependencies, not chronology. Incoming arrows are used jointly; transitive dependencies are usually omitted. Detailed input and external-theorem nodes appear on pages 2–5.

All construction-dependent conclusions inherit the stated finite inputs. Number/page tags refer to `hi_cft_c8_proof.pdf` (78 pages). This is a map of the draft, not additional proof certification.

Finite construction and the all-weight argument

Appendices A–E: actual-vector provenance comes before closure, and properness comes before character identification.

Finite input | Argument in draft | Conclusion | External input | Optional route | Open boundary



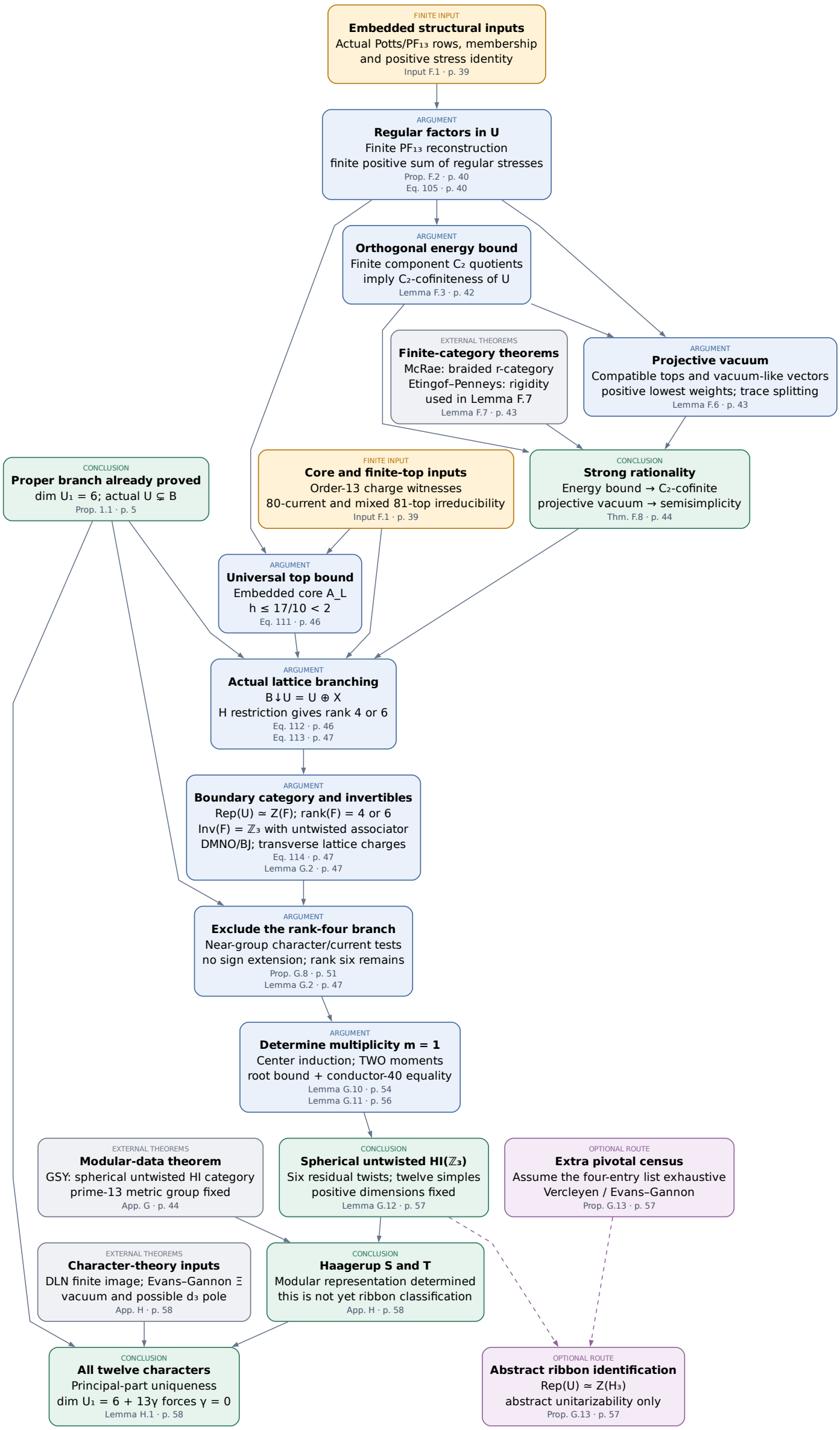
The projected residuals alone do not prove Gate 4: raising premises and saturation ranks are also used. Gate 5 and its positive-mode consequences are derived; the value $\dim U_4 = 6010$ is not an input.

All construction-dependent conclusions inherit the stated finite inputs. Number/page tags refer to `hi_cft_c8_proof.pdf` (78 pages). This is a map of the draft, not additional proof certification.

Rationality, branching, and characters

Appendices F–H: the character proof uses modular data, not an assumed tensor-category identification.

Finite input | Argument in draft | Conclusion | External input | Optional route | Open boundary



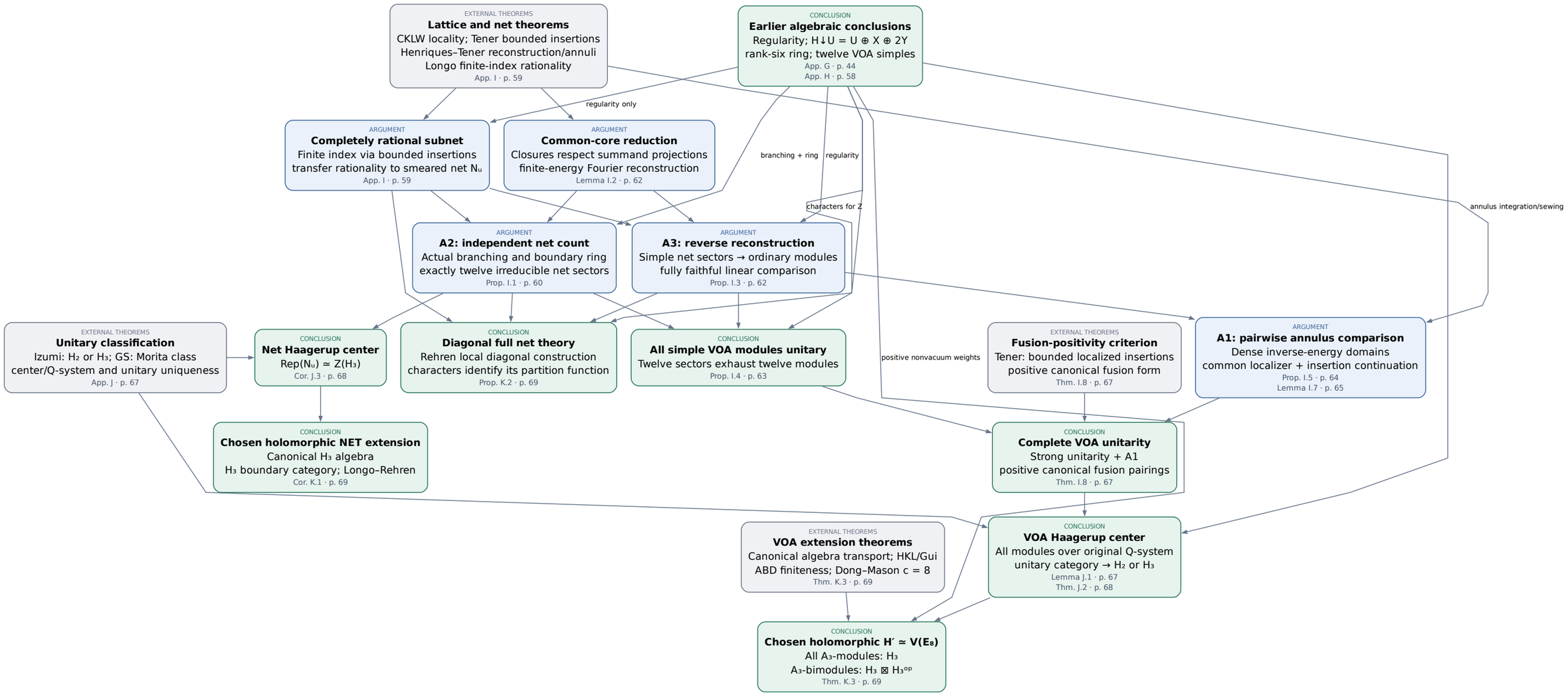
Dashed arrows mark an optional route using the extra pivotal-classification hypothesis. It is not used for characters or the main analytic realization, and it does not establish canonical VOA fusion positivity.

All construction-dependent conclusions inherit the stated finite inputs. Number/page tags refer to `hi_cft_c8_proof.pdf` (78 pages). This is a map of the draft, not additional proof certification.

Analytic realization: two distinct routes

Appendices I–K: positivity of the vacuum, of all modules, and of fusion pairings are separate assertions.

Finite input | Argument in draft | Conclusion | External input | Optional route | Open boundary



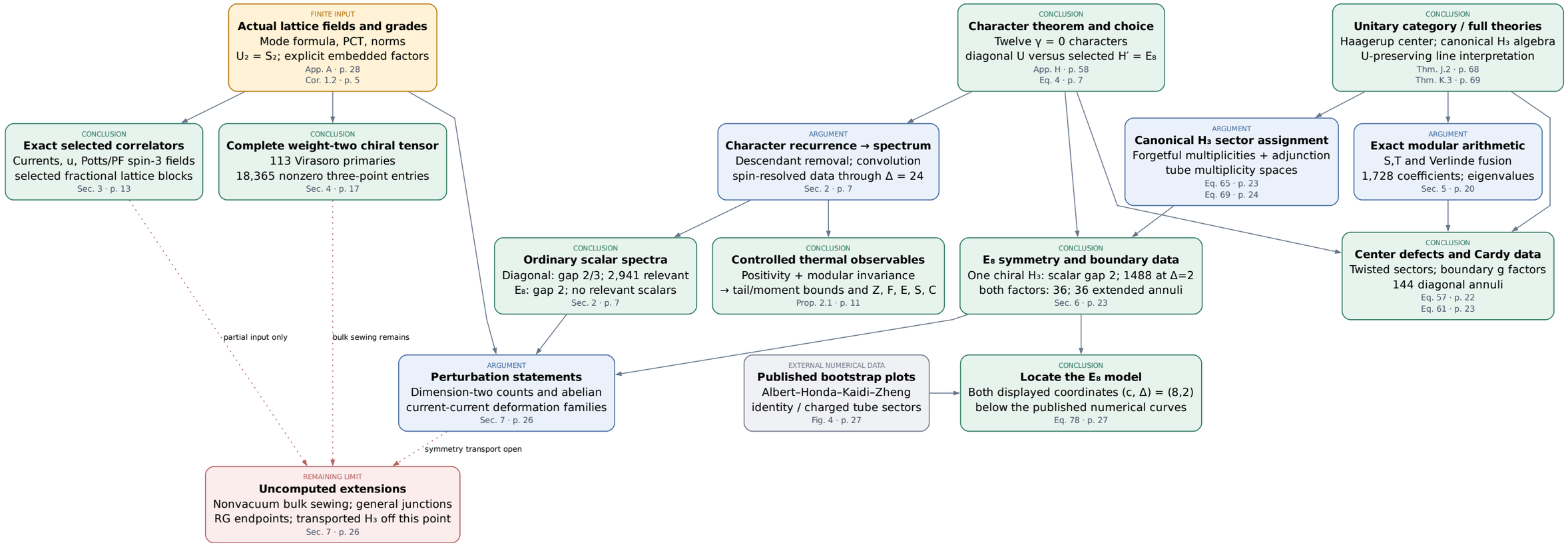
A2 and A3 share core methods but are parallel arguments. The net route needs no A1; the chosen holomorphic NET is not identified with E8. The chosen VOA extension H' need not equal the original embedding H/U. Haagerup categories use all algebra modules, not only local modules.

All construction-dependent conclusions inherit the stated finite inputs. Number/page tags refer to `hi_cft_c8_proof.pdf` (78 pages). This is a map of the draft, not additional proof certification.

From the construction to physical observables

Sections 2–7: numerical datasets have explicit mathematical inputs and two different full-theory interpretations.

Finite input | Argument in draft | Conclusion | External input | Optional route | Open boundary



The bootstrap point uses c = cL = cR = 8, not 16. Diagram endpoints record what the draft computes; they do not assert a complete bulk OPE, a new certified SDP exclusion bound, or a symmetry-preserving conformal manifold. All construction-dependent conclusions inherit the stated finite inputs. Number/page tags refer to `hi_cft_c8_proof.pdf` (78 pages). This is a map of the draft, not additional proof certification.